

Modeling Insurance Risk with Self-Exciting Processes

Why Standard Poisson Models Underestimate Ruin Probability

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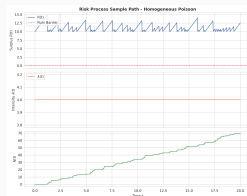
Standard models underestimate the risk of contagion

The Status Quo (Poisson)

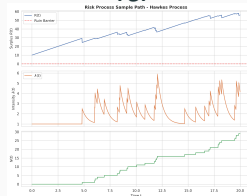
- Assumes claims are **independent**.
- Good for: Car accidents, house fires.
- Fails for:** Epidemics, natural disasters, chain collisions.

The Reality

- Events cluster in time.
- One claim increases the probability of the next.



VS.



The Cramér-Lundberg model must adapt to history

The classical surplus equation:

$$R(t) = u + ct - \sum_{i=1}^{N(t)} X_i$$

- u : Initial Capital (Reserves).
- c : Premium Rate (Income).
- $N(t)$: **The weak link.**

If $N(t)$ ignores history, we misprice the risk of ruin.

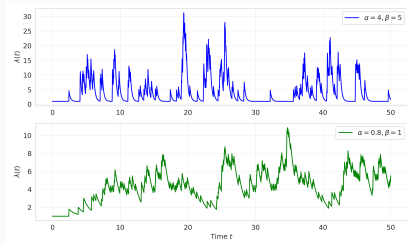
Hawkes processes capture the "Aftershock" effect

The intensity $\lambda(t)$ is not constant; it reacts to history.

$$\lambda(t) = \mu + \sum_{t_i < t} \alpha e^{-\beta(t-t_i)}$$

Key Mechanics:

- μ : Baseline risk (day-to-day).
- **Jump** (α): Sudden panic/contagion.
- **Decay** (β): System recovery speed.



Average profitability does not guarantee solvency

The Net Profit Condition (NPC)

$$c > \frac{\mu}{1 - n} \cdot \mathbb{E}[X]$$

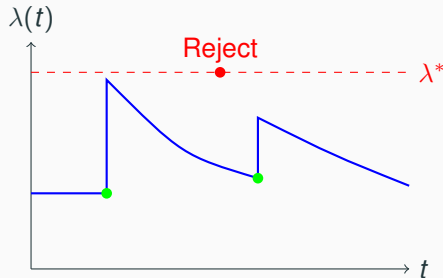
- **The Trap:** Two models can have the **same average** number of claims but vastly different risks.
- **The Danger:** Stability depends on the branching ratio $n = \alpha/\beta < 1$.
- Meeting the NPC is **necessary, but not sufficient**.

We use Ogata's Algorithm to simulate path-dependence

Since risk changes every second, standard simulations fail.

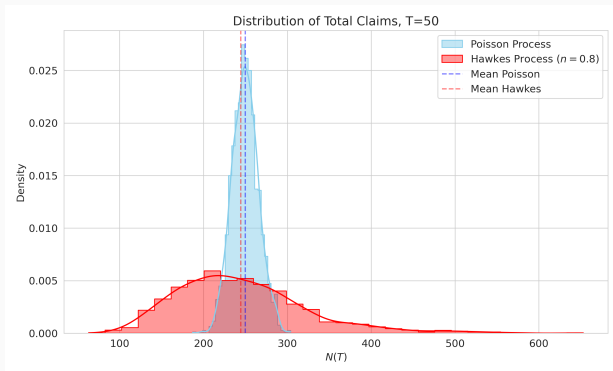
Methodology:

1. Simulate a "worst-case" rate λ^* .
2. Rejection sampling based on history.
3. Exact simulation (no bias).



Clustering creates dangerous "Fat Tails"

Comparing Poisson vs. Hawkes with the **same mean** ($\mathbb{E}[N] = 250$).

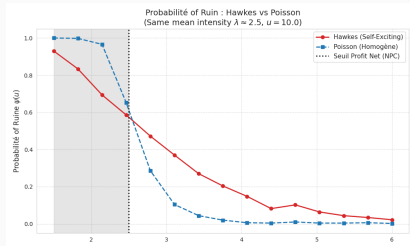


- **Poisson**: Predictable, bell-shaped.
- **Hawkes**: High probability of **extreme scenarios**.

Hawkes models reveal hidden bankruptcy risks

Probability of Ruin $\psi(u)$ vs. Premium c .

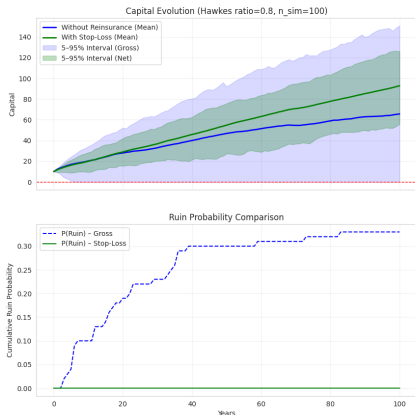
- **Solvent Zone** ($c > c_{min}$):
- Poisson model predicts safety (Blue).
- Hawkes model predicts **ruin** (Red).
- **Result:** Reserves calculated on Poisson are insufficient.



Stop-Loss Reinsurance mitigates the clustering risk

We transfer the "Tail Risk" to a reinsurer.

$$I_t = \min(S_t, P)$$



Outcome:

- **Blue (Gross)**: High volatility, ruin probable.
- **Green (Net)**: Stable growth, ruin eliminated.

Recommendation

Conclusion

1. **Acknowledge Contagion:** Standard models dangerously underestimate variance.
2. **Monitor Parameters:** α (Jump) and β (Memory) are as important as the mean.
3. **Purchase Protection:** Capital reserves alone are inefficient; Stop-Loss reinsurance is required for stability.

Do you have any questions?

References i

-  Hawkes, A. G., & Oakes, D. (1974). A cluster process representation of a self-exciting process. *Journal of Applied Probability*.
-  Bacry, E., et al. (2015). Hawkes processes in finance. *Market Microstructure and Liquidity*.
-  Ogata, Y. (1981). On Lewis' simulation method for point processes. *IEEE Transactions*.